

Riley Beenders

ENGINEERING LEADERSHIP | SOFTWARE-INTEGRATED SYSTEMS | TECHNOLOGY STRATEGY

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PROFESSIONAL SUMMARY

Engineering and technology leader with 10+ years of multidisciplinary experience, including CTO and Lead R&D roles spanning hiring, vendor partnerships, and technology strategy. Delivers software-integrated products and internal business systems, including a self-developed ERP/CRM on a redundant Kubernetes cluster, remote operations and analytics platforms, and Python, LabVIEW, PowerShell, and TypeScript/Node.js applications. Tracks emerging technology, including AI-assisted diagnostics and local LLM tooling, and turns it into practical, reliable, well-supported systems.

CORE EXPERTISE

Leadership & Delivery: Technical team leadership; hiring and technical interviews; external vendor and software-partner management; vendor cost negotiation; technology strategy; requirements definition; worldwide collaboration

Software: Python; TypeScript/React/Next.js (Node.js runtime); LabVIEW; PowerShell; Git/GitHub; ERP/CRM application development; Modbus data integration; application testing and debugging

Platforms & Reliability: Kubernetes; Docker; Proxmox virtualization; Windows/Linux administration; Vercel deployment; GitHub Actions; redundant self-hosted services; metrics, live monitoring, and web analytics

Emerging Tech & Operations: AI-assisted diagnostics and machine-vision alerts; local LLM tooling (LM Studio, oMLX); Microsoft Business Central ERP; process standardization; risk analysis; technical documentation

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.
- Implemented ProteorPrint OS with external software partner SimplyPrint, contributing machine requirements and integration for remote operation, live video, AI-assisted diagnostics, and material, waste, and outcome metrics.
- Managed the Coplay site's Microsoft Business Central ERP rollout, syncing SKUs with Proteor's global locations.
- Managed and expanded the division's IT infrastructure after the Proteor acquisition, deploying cloud gateways and standardizing software and internal processes to align with company-wide systems.
- Built an automated LabVIEW QC system that negotiates Modbus communication with three tri-axial laser sensors over a dedicated private VLAN, collecting nine data points (three per laser) every 17.5 ms and charting live tolerances.
- Automated QC reporting: each run generates a formatted PDF stored on a remote server under an ID that becomes a QR code on the product, making field issues traceable.
- Led the design and development of ICARUS-Lite, owning its mechanical, electrical, and software design, including custom AC heaters and an IEEE-style electrical design adaptable to 120/240 VAC.
- Owned ICARUS-Lite through commercialization with plug-and-play setup, contributing to increased revenue.
- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.
- Provide cross-functional technical support for the product line, from troubleshooting to after-sales service.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer

January 2023 - January 2025

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while retaining hands-on responsibility for mechanical, electrical, firmware, and software systems.
- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM for orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation linked to purchasing.

- Led the design and launch of Gen3 POSEIDON, combining a 1 m³ build volume, filament-and-pellet dual extrusion, braked NEMA 34 Z drives that prevent uncontrolled motion during power loss, industrial-servo XY motion, and a custom gantry.
- Researched high-speed linear motors that moved a 30 kg payload at 3 m/s with 5-micron accuracy and conducted early 6-axis robotic non-planar printing research.

Lead Electro-Mechanical Engineer

June 2022 - January 2023

- Developed independently testable electronics enclosures that could be installed in minutes and coordinated custom harnesses, panel designs, and hardware improvements with vendors.
- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging while supporting larger, modular machines.

Industrial Engineer

April 2021 - June 2022

- Designed THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry system for a large-scale statue client, and managed CAD, BOM, sourcing, fabrication, assembly, firmware tuning, support, and training.

Engineering Technician

May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized process parameters while documenting design limitations and opportunities for product improvement.

SELECTED PROJECTS

RileyBeenders.com - Full-Stack Portfolio Platform | 2026 | Built and deployed a TypeScript/React/Next.js site on Vercel with structured JSON content, a Node.js PDF endpoint, Git version control, GitHub Actions automation on every push to main, and web analytics.

Engineering Operations Platform - Kubernetes ERP/CRM | Filament Innovations | Redundant Kubernetes-hosted system connecting engineering design data to operations: centralized orders, inventory, tasks, projects, and customer interactions, and linked dynamic BOM generation to purchasing.

PROTEOR Print Integration - Remote Operations Software | Proteor | Single interface for remote operation and diagnosis from any networked device, delivered with an external software partner: live video monitoring, machine-vision AI-assisted failure detection and alerts, and tracking of material usage, waste, and print success or failure for analytics-driven decisions.

RB's Auto App Updater - PowerShell / Windows Automation | 2026 | One-line PowerShell install that upgrades every WinGet package through an elevated scheduled task: a non-elevated launcher calls protected worker scripts, admin approval is requested once at setup, per-package failures never stop the run, exit codes map to plain-language statuses, and self-updates install only after SHA-256 verification against a version manifest.

ExtrusionLine Studio - LabVIEW Metrology | Proteor | Replaced a manual measurement workflow with automated Modbus acquisition from three tri-axial laser sensors on a private VLAN, hands-free startup, live tolerance charts, and QR-traceable PDF reports on a remote server.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed 750+ wireless access points across six school buildings; automated setup-data entry for 2,500+ Chromebooks with an Arduino-based tool; coordinated distribution with school administrators.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and operate a small online store for printer heat inserts and accessories; produced plastic face shields during the COVID-19 pandemic.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Certifications: MathWorks MATLAB, Simulink, Simscape, and Machine Learning Onramps (2026); Lean Six Sigma Yellow Belt and White Belt, Educate 360 (2025); Penn State Certificate in Engineering Design and Tools (2023); Lafayette College Mechanical Engineering Certificate (2019)